

# Scientific Machine Learning: Blending Science with Data

## Homework Days 10+11

- a) Download and plot the data from the file ‘DataHomeworkDay11.csv’
- b) **(LEVEL 1)** Write a program in Julia that finds the best fitting polynomial to the data in this file.
- c) **(LEVEL 2)** Try to fit the data to a Neural Network with 1 hidden layer and one node. You must use the  $\tanh()$  activation function between in the input and the hidden layer. Can you find a good fit? Can you explain why?
- d) **(LEVEL 3)** Write a Julia program that uses a Neural Network to find a fit to this data set.